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EFFECT OF DIFFERENT LAND-USE TYPES ON THE HUMUS STATUS AND AGROCHEMICAL PROPERTIES OF IRRIGATED LIGHT SEROZEM SOILS IN THE ARID ZONE OF KAZAKHSTAN (A CASE STUDY OF THE KYZYLORDA REGION)

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ABSTRACT: Light sierozems are widely used for irrigated agriculture in southern Kazakhstan, but their low organic matter content and susceptibility to secondary salinization make soil fertility sensitive to land use. This study compared the humus status and agrochemical properties of irrigated light sierozems under contrasting land-use types in the Kyzylorda Region. Tree soil profiles were studied within a third-year alfalfa field, irrigated cropland under melon production, and land abandoned for more than 15 years. Samples were collected by genetic horizon and analyzed for particle-size composition, soil organic carbon (SOC), humus fractions, available nutrients, pH, and exchangeable cations. All soils had a light loam texture dominated by fine sand. Surface SOC content was 0.96% under alfalfa, 0.72% in irrigated cropland, and 0.65% in the fallow soil. The alfalfa soil contained more non-hydrolysable residue and humic acids. The $\Sigma\text{HA}/\Sigma\text{FA}$ ratio was 0.6 in all soils. Nitrate nitrogen and available sulfur accumulated in the middle horizons under alfalfa and irrigated cropland, whereas available phosphorus and exchangeable potassium were concentrated mainly in the upper horizons. Soil pH ranged from 8.76 to 9.07. The soil under third-year alfalfa had the highest SOC content and a greater contribution of stable humus fractions. At the studied site, more than 15 years of unmanaged fallow did not restore a comparable stable organic matter pool.

KEYWORDS: light sierozem; land-use type; irrigation; soil organic carbon; humic substances; nutrients.

CONFLICT OF INTEREST: The authors declare that there is no conflict of interest.

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INTRODUCTION

The rational use of land resources and the preservation of soil fertility are crucial for the sustainable functioning of irrigated agriculture. For the arid zone of southern Kazakhstan, this issue is especially relevant because the combination of intensive agricultural use and natural characteristics of the area leads to accelerated mineralization of organic matter, deterioration of soil physical and chemical properties, and development of secondary salinization processes.

Light sierozem soils are one of the main soil resources in the arid zone of southern Kazakhstan. They develop under conditions of high heat availability and insufficient rainfall, which determine their specific properties. Light sierozem soils are characterized by a high level of biological activity, intensive organic matter mineralization, and relatively low humus content (Soils of the Kazakh SSR, 1988). As a result, light sierozem soils show high variability in humus levels and agrochemical properties depending on the type of land use.

For the Kyzylorda Region, soil fertility preservation is becoming especially important. The results of long-term studies confirm the conclusion that in irrigated sierozem soils, agricultural use leads to a 40-50% decrease in humus content and mobile forms of nutrients over a relatively short period (Suleimenov et al., 2013). At the same time, under irrigation poor drainage, rising groundwater levels, and high evaporation can increase the capillary rise of salts and development of secondary soil salinization (Suleimenov et al., 2009).

Along with changes in water-salt balance, one of the most vulnerable components of irrigated sierozem soils is organic matter, including humus, is one of the most vulnerable components of irrigated sierozem soils which are considered the key components of soil fertility and determine soil stability. Therefore, for light sierozem soils, even small changes in the direction of humus formation and humus content can noticeably affect water retention, buffering capacity, and the ability to maintain productive potential of agroecosystems. This is especially important for light sierozem soils with a low initial humus content. Even minor quantitative and qualitative changes in organic matter significantly influence the resilience of agrocenoses to unfavorable factors (Mamontov et al., 2023).

In irrigated agriculture, when there are risks of secondary salinization, both the availability of soil nutrients and the physical and chemical properties of the soils are important, which are largely determined by the state of the soil exchange complex. Changes in the balance of exchangeable cations, primarily calcium, magnesium, and sodium, can significantly affect soil structure, its water-physical properties, pH level, and direction of humus formation processes (Orlov, 1992; Vorobyeva, 1998).

Despite a significant number of studies on the fertility of sierozems, most of them focus on specific indicators of their condition. At the same time, comparative assessment of humus status, nutrient availability, and cation exchange composition of light sierozems under different land use types has been insufficiently studied. There are practically no data comparing the properties of irrigated arable land, perennial legume grasses, and long-term fallow in the arid zone of Kazakhstan.

In this regard, the aim of the study was to comparatively assess humus status, agrochemical properties, and composition of exchangeable cations in light sierozems under different types of land use in the arid zone of Kazakhstan, using the Kyzylorda region as an example. Special attention was paid to a comprehensive study of organic carbon content, the group and fractional composition of humus, soil fertility in terms of available nutrients, and patterns of exchangeable cation distribution in the soil profile under irrigated arable land, alfalfa crops, and long-term fallow. This made it possible to carry out a comprehensive comparative assessment and clarify specific features of functioning of light sierozems in the arid zone of Kazakhstan.

MATERIALS AND METHODS

The light sierozem soils of the Karatau foothill plain are located southeast of the village of Zhanakorgan at an elevation of 169–170 meters above sea level and have formed on Quaternary

deluvial-proluvial deposits. The groundwater is mainly fresh, occurs at a depth of 8–12 meters, and belongs to the bicarbonate-calcium type. With prolonged irrigation and changes in hydrogeological conditions, the groundwater level may rise, affecting the soil water-salt balance.

For the study, irrigated light sierozems were selected within the “Baldyrgan” farm in Zhanakorgan district, Kyzylorda Region (Fig. 1).



Figure 1. Locations of the soil profile sites

To comparatively assess soil properties under different types of land use, three soil profiles were established within an area of approximately 20 hectares:

- profile No. 1 – third-year alfalfa field (43.966422° N, 67.230940° E);
- profile No. 2 – irrigated arable land under melon crops (43.967021° N, 67.230905° E);
- profile No. 3 – fallow land withdrawn from agricultural use more than 15 years ago (43.967349° N, 67.229111° E).

The preceding crop in the alfalfa fields was watermelon; in the irrigated cropland, melon crops were sown after three years of alfalfa; before the fallow site was withdrawn from agricultural use, it was cultivated with melon crops and potatoes. After the land was withdrawn from agricultural use, it was not cultivated for more than 15 years, and its present vegetation consist mainly of saxaul and xerophytic thorny plants.

The research was carried out using soil-profile and morphological methods, as well as a set of laboratory agrochemical and physicochemical methods. Soil properties were studied by genetic horizons throughout each profile. Individual soil samples (approximately 500 g) were collected from each genetic horizon using the continuous sampling method for laboratory analysis. Three replicate soil samples were collected from each genetic horizon.

Soil laboratory analyses were carried out using standard methods:

Soil sampling	GOST 58595–2019				
Sample preparation for analysis	GOST 29269–91				
Particle-size distribution	pipette method (ISO 11277:2020)				
Soil pH in a water suspension	ISO 10390:2021, potentiometric method				
Organic carbon, %	GOST	26213-2021	“Soils.	Method	for

	determination of organic matter”
Nitrate nitrogen, mg/kg	GOST 26951-86 “Soils. Determination of nitrates by ionometric method”
Available phosphorus and exchangeable potassium	GOST 26205-91 “Soils. Determination of available phosphorus and potassium compounds by the Machigin method as modified by CISAA”
Exchangeable cations: Ca, Mg, and Na	GOST 26487-85 “Soils. Determination of exchangeable calcium and exchangeable (mobile) magnesium by CISAA methods”; GOST 26950-86 “Soils. Method for determination of exchangeable sodium”
Available S	GOST 26490-85 “Soils. Determination of mobile sulfur by the CISAA method”
Group and fractional composition of humus: humic acids (HA) fractions I–III; fulvic acids (FA) fractions I–III; non-hydrolysable residue (NHR, humin).	I. V. Tyurin (modified by V. V. Ponomareva and T. A. Plotnikova).

Statistical analyses were performed using Microsoft Excel 2021. Descriptive statistics were calculated for all measured variables. Relationships among soil physical and agrochemical properties were evaluated using Spearman's rank correlation coefficient (ρ). Correlation matrices were used to identify associations between particle-size distribution, soil organic matter characteristics, nutrient availability, and exchangeable cations.

RESULTS

The results of particle-size analysis showed that all the studied light sierozem soils had a light loam texture and were characterized by the predominance of the fine-sand fraction (0.25–0.05 mm), which ranged from 57.71 to 75.72% (Table 1).

Table 1. Particle-size composition of light sierozem soils under different land use types

Land use types	Depth, cm	Gravel >1 mm	Coarse sand, 1.0–0.25 mm	Fine sand, 0.25–0.05 mm	Coarse silt, 0.05–0.01 mm	Medium and fine silt, 0.01–0.001 mm	Clay, <0.001 mm	Physical clay, <0.01 mm
Alfalfa	0–35	1.16	0.13	61.78	24.69	12.24	1.16	13.40
	35–50	0.43	0.10	61.96	25.39	12.12	0.43	12.55
	50–70	0.14	0.16	60.71	27.35	11.64	0.14	11.78
	70–100	0.03	0.05	57.71	30.59	11.63	0.03	11.66
Arable land	0–33	0.20	0.10	63.73	24.02	11.95	0.20	12.15
	33–48	0.19	0.06	59.56	28.87	11.30	0.20	11.50
	48–72	0.03	0.04	66.02	25.43	8.49	0.03	8.52
	72–100	0.04	0.07	75.72	15.61	8.57	0.04	8.61
Fallow land	0–30	0.27	0.25	69.77	16.63	13.08	0.27	13.35
	30–43	0.33	0.30	68.89	16.03	14.45	0.32	14.77
	43–67	0.32	0.34	71.37	15.40	12.57	0.32	12.89
	67–100	0.45	0.35	72.92	15.72	10.56	0.45	11.01

The proportion of coarse fractions (>0.25 mm) in all horizons was insignificant and did not exceed 1.3%. The most even distribution of the fine-sand fraction throughout the profile was noted in soil under alfalfa, where its content ranged from 61.78 to 61.96% in the upper part of the profile and decreased to 57.71% in the 70–100 cm horizon. In the fallow area, the content of this fraction also changed slightly, gradually increasing with depth from 68.89–69.77% to 72.92%. The most pronounced profile differentiation was observed in irrigated arable land, where the content of the fine-sand fraction increased from 63.73% in the plough horizon to 75.72% in the lower part of the profile.

The content of physical clay (<0.01 mm) in the studied soils studied ranged from 8.52 to 14.77%. The highest values were observed in the fallow soil (13.35–14.77%), while the irrigated arable soil had the lowest physical clay content (8.52–12.15%). Soil under perennial alfalfa occupied an intermediate position, with physical clay content ranging from 11.66 to 13.40%.

The evaluating of the humus status of light sierozem soils in the topsoil showed significant differences among the studied land-use type. The highest soil organic carbon (SOC) content was recorded under perennial alfalfa (0.96%), whereas it was 0.72% in irrigated arable and 0.65% in fallow land, corresponding to decreases of 25.0 and 32.3% relative to the alfalfa soil, respectively (Fig. 2a).

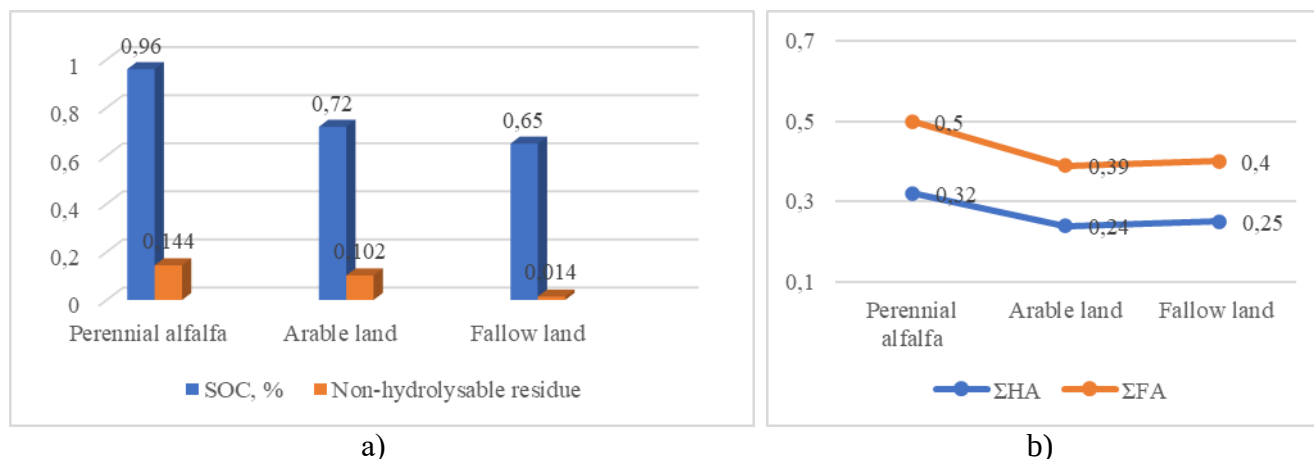


Figure 2. Organic matter content and composition in light sierozem soils under different land-use types: (a) soil organic carbon (SOC), % and non-hydrolysable residue; (b) carbon content of individual humic substance fractions (% of soil C).

The content of non-hydrolysable residue also varied among the studied land-use types. In soil under alfalfa, this value was 0.144% C, compared with 0.102% C in irrigated arable land and 0.014% C in the fallow soil (Fig. 2a).

The analysis of the group and fraction composition of humus showed that the soil under third-year alfalfa differed from the other sites not only in organic carbon content but also in the qualitative composition of humus substances. In all variants, fulvic acids predominated in the composition of humic substances composition (Fig. 2b).

The total fulvic acid content was 0.50% C in the alfalfa soil and 0.40% C in the fallow soil, compared 0.39% C in the irrigated arable soil. The distribution pattern of humic substance fractions largely depended on land use type. The highest total content of humic acids was observed in the plot with perennial alfalfa at 0.32% C, while in arable and fallow land, it was 0.24% and 0.25%, respectively (Table 2).

Table 2. Humus fractional composition of light sierozem soils under different land-use types

Land use types	HA*			FA*			C _{total} (HA+FA)	C _{ha} /C _{fa}
	I	II	III	I	II	III		
Alfalfa	0.08	0.03	0.21	0.14	0.09	0.27	0.820	0.6
	9.8	3.31	26.1	16.7	11.4	32.7		
Arable land	0.05	0.08	0.11	0.15	0.08	0.16	0.630	0.6
	8.58	12.9	17.3	23.5	12.3	25.4		
Fallow land	0.07	0.05	0.13	0.22	0.06	0.12	0.650	0.6
	11.2	7.34	21.0	34.1	9.43	18.1		

* The numerator shows carbon content of individual fractions of humic substances (% C of soil), and denominator – their share in the total carbon content of the isolated HA and FA fractions (%).

The highest total carbon content of isolated fractions of humic substances was found in soil under alfalfa – 0.820%, while in arable land and fallow plot it was 0.630% and 0.650%, respectively.

The ratio $\Sigma\text{HA}/\Sigma\text{FA}$ was 0.6 in all cases.

In soils studied, the HA_{III} fraction accounted for 17.3% of total SOC in the arable soil and 26.1% in the soil under alfalfa.

In the alfalfa and arable soils, the FA_{III} fraction predominated, accounting for 32.7% and 25.4%, respectively. In soil of fallow plot, among the fulvic acids, FA_{I} fraction predominated – 34.1%, while the share of FA_{III} was 18.1%.

All the studies soils were characterized by a low nitrate nitrogen content in the topsoil (no more than 10 mg/kg). Its accumulation in the middle part of the profile was observed only under alfalfa and in irrigated arable land (Table 3).

Table 3. Agrochemical properties of light sierozem soils under different land-use types

Land use types	Depth, cm	N-NO ₃ , mg/kg	P ₂ O ₅ , mg/kg	K ₂ O, mg/kg	S avail, mg/kg	pH
Alfalfa	0–35	9.33 ± 0.25	26.77 ± 3.68	304.53 ± 3.76	13.95 ± 2.53	8.78 ± 0.01
	35–50	11.93 ± 0.50	16.37 ± 2.17	257.60 ± 1.83	22.78 ± 2.36	8.95 ± 0.01
	50–70	41.70 ± 0.40	15.57 ± 2.48	225.76 ± 4.73	55.67 ± 0.20	8.82 ± 0.01
	70–100	11.56 ± 0.25	8.32 ± 1.51	187.43 ± 2.91	26.73 ± 1.66	9.05 ± 0.01
Arable land	0–33	3.00 ± 0.17	36.77 ± 1.79	397.10 ± 1.83	5.59 ± 3.37	8.99 ± 0.01
	33–48	3.36 ± 0.15	9.43 ± 0.09	183.16 ± 4.73	14.28 ± 2.18	8.98 ± 0.01
	48–72	27.86 ± 0.28	3.55 ± 1.40	170.83 ± 4.10	76.88 ± 0.20	8.76 ± 0.01
	72–100	13.50 ± 0.20	3.52 ± 1.88	132.26 ± 3.98	57.54 ± 2.73	8.85 ± 0.01
Fallow land	0–30	2.13 ± 0.20	54.44 ± 1.97	267.60 ± 5.64	3.72 ± 1.76	9.00 ± 0.01
	30–43	1.45 ± 0.10	31.84 ± 3.94	171.46 ± 4.48	5.06 ± 2.85	9.07 ± 0.02
	43–67	2.63 ± 0.20	16.21 ± 4.05	186.63 ± 6.33	8.20 ± 0.73	9.05 ± 0.01
	67–100	10.26 ± 0.25	14.65 ± 0.11	231.03 ± 0.01	13.95 ± 2.53	8.78 ± 0.01

Under alfalfa and in the arable plot, the maximum nitrate nitrogen content was recorded in the middle part of the profile and amounted to 41.70 and 27.86 mg/kg, respectively.

Unlike nitrate nitrates, available phosphorus was concentrated in the upper horizons, where is content was 26.77 mg/kg under alfalfa, 36.77 mg/kg in the arable soil, and 54.44 mg/kg in the fallow soil.

The content of exchangeable potassium was also highest at the top of profile. The level of exchangeable potassium in soil ranges from medium (fallow) to high (arable land, alfalfa) – 268–397 mg/kg, with further decrease deeper in profile.

The analysis of distribution of available sulfur in soil profile showed significant differences between the studied land use types. In the upper horizons of fallow plot and arable land, the available sulfur content was low, amounting to 3.72 and 5.59 mg/kg of soil, respectively. Under perennial alfalfa, the available sulfur content corresponded to a medium level of availability and amounted to 13.95 mg/kg.

As shown in Table 3, irrigated light sierozems, regardless of land-use type, were characterized by a strongly alkaline reaction, with pH values ranging from 8.76 to 9.07.

In all the studied variants, calcium and magnesium predominated in composition of exchangeable bases, while the content of exchangeable sodium remained low (Table 4).

Table 4. Exchangeable cation content in light sierozem soils under different land-use types, mmol (+)/100 g soil

Land use types	Depth, cm	Ca ²⁺	Mg ²⁺	Na ⁺
1	2	3	4	5
Alfalfa	0–35	10.203 ± 0.006	4.523 ± 0.003	0.163 ± 0.015

	35–50	6.297 ± 0.006	4.103 ± 0.006	0.183 ± 0.015
	50–70	6.403 ± 0.006	3.523 ± 0.003	0.280 ± 0.010
	70–100	5.747 ± 0.006	5.203 ± 0.006	0.160 ± 0.010
1	2	3	4	5
Arable land	0–33	6.297 ± 0.006	2.980 ± 0.009	0.113 ± 0.006
	33–48	7.373 ± 0.003	3.730 ± 0.009	0.120 ± 0.010
	48–72	9.573 ± 0.003	5.023 ± 0.003	0.150 ± 0.010
	72–100	7.247 ± 0.006	5.347 ± 0.006	0.120 ± 0.010
Fallow land	0–30	6.647 ± 0.006	6.423 ± 0.003	0.150 ± 0.010
	30–43	6.930 ± 0.009	5.430 ± 0.009	0.120 ± 0.010
	43–67	16.580 ± 0.009	7.903 ± 0.006	0.123 ± 0.006
	67–100	19.647 ± 0.049	7.903 ± 0.006	0.167 ± 0.012

In the top horizon, the highest content of exchangeable calcium was noted under perennial alfalfa, while in soils of other plots, its content was 1.5–1.6 times lower. The fallow soil was characterized by increase in magnesium content in the upper part of profile up to 6.423. The irrigated arable land had the lowest values of Ca^{2+} and Mg^{2+} cations – 6.297 and 2.980 mmol (+) / 100 g of soil in the top part of profile.

The profile distribution of exchangeable bases depended on land use type. Under alfalfa, calcium content generally decreased with depth; in arable land, it increased in the middle part of profile; and on fallow land, it sharply increased in the lower horizons, reaching 19.647 mmol (+) per 100 g of soil.

Magnesium content changed less noticeably, but its accumulation in the lower part of profile was also most pronounced on fallow land. Exchangeable sodium was present in small amounts in all cases and did not exceed 0.280 mmol/100 g, indicating its low contribution to the soil exchange complex.

For a more detailed analysis of the distribution of agrochemical indicators across the soil profiles and the relationships among them, correlation analysis was carried out using Spearman's rank correlation coefficient for all sampled horizons ($n=12$).

It was found that the content of nitrate nitrogen had strong positive correlation with available sulfur ($\rho = 0.93$; $p < 0.001$) and negative correlation with pH ($\rho = -0.75$; $p = 0.0046$) and available phosphorus ($\rho = -0.61$; $p = 0.036$). This reflects differences in profile distribution and behavior of soil nutrients.

Available phosphorus is positively correlated with exchangeable potassium ($\rho = 0.80$; $p = 0.0016$) and proportion of “physical clay” (<0.01 mm) ($\rho = 0.77$; $p = 0.0034$), highlighting the role of fine fractions and topsoil layers in accumulation and retention of P and K. In contrast, available sulfur is negatively associated with “physical clay” ($\rho = -0.84$; $p = 0.0006$) and available phosphorus ($\rho = -0.79$; $p = 0.0022$), which aligns with its higher mobility and accumulation in sub plow/lower horizons. The Ca and Mg correlation ($\rho = 0.75$; $p = 0.0052$) reflects coordinated changes in composition of absorbed cations across the studied profiles.

Additionally, Spearman's rank correlation coefficients (ρ) were calculated for the topsoil horizons (0–30/35 cm), including soil organic carbon and humus fractions (ΣHA and ΣFA). (Fig. 3).

Due to the small number of observations ($n = 3$), these coefficients were used only for a tentative assessment of the direction of the relationships and were not interpreted as statistically significant.

DISCUSSION

The analysis of changes in particle size distribution of soils showed that observed differences are likely due to combined influence of land use features and natural spatial variability of the particle size distribution of the studied light sierozem soils.

The obtained data align with the known characteristics of light-textured, light-colored sierozems, which are distinguished by high water permeability, relatively low water-holding capacity, and limited sorption capacity.

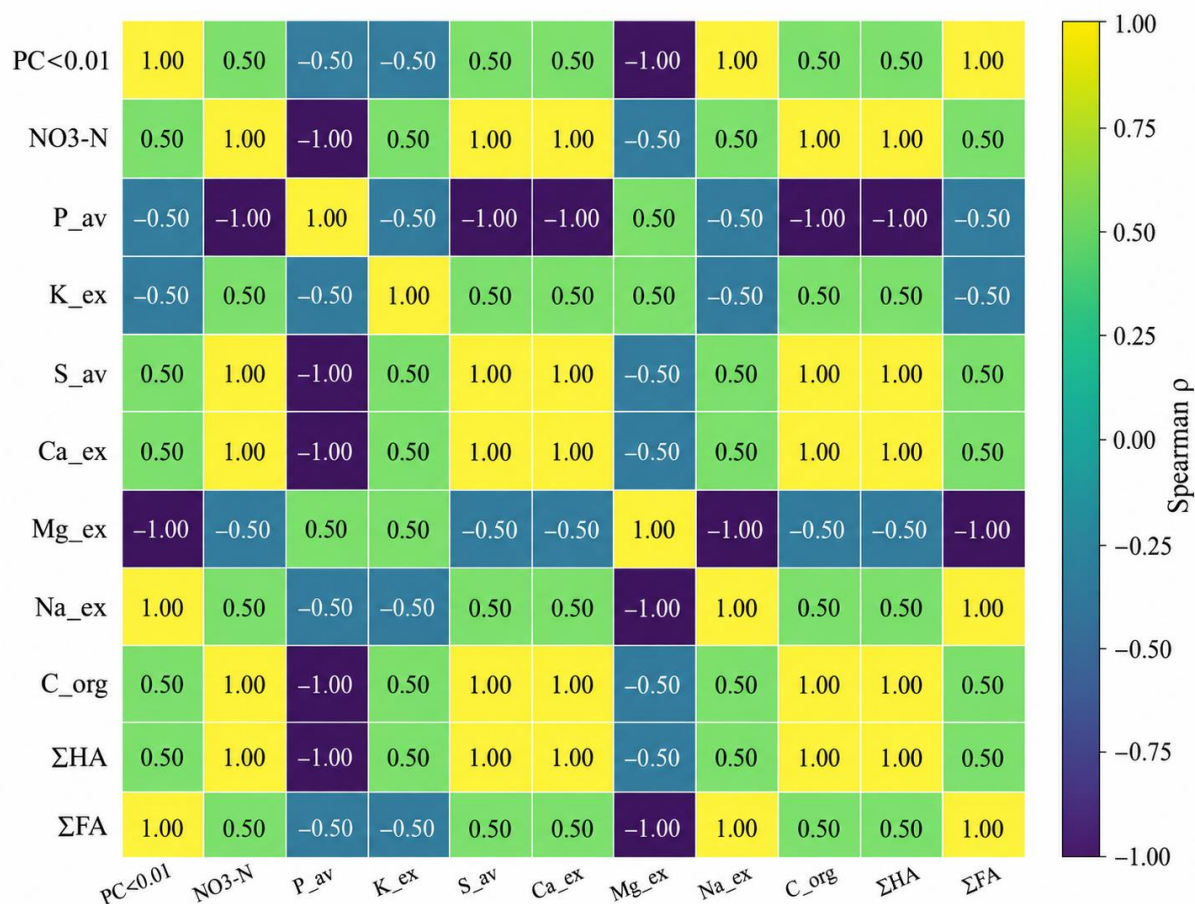


Figure 3. Matrix of Spearman's rank correlation coefficients (ρ) for 0–30/0–35 cm soil layer

Under these conditions, the most mobile compounds—primarily nitrates and sulfates – undergo easier redistribution throughout soil profile, whereas phosphorus and significant portion of potassium concentrate predominantly in the upper horizons. The predominance of fine-sand fraction, combined with low content of silt-sized particles, reflects formation characteristics of the studied light-colored sierozems on Quaternary deluvial-proluvial deposits (Borovsky et al., 1971).

The results of assessing humus condition of light sierozem soils are consistent with numerous data on positive effect of long-term legume crops on soil organic matter accumulation. The higher organic carbon content in the soil under third-year alfalfa may be associated with its well-developed root system and annual input of root and crop residues.

Symbiotic nitrogen fixation also plays important role, ensuring formation of plant biomass with higher nitrogen content and creating favorable conditions for microbial transformation of plant residues and subsequent humus formation (Zhou et al., 2015; Jensen et al., 2012; Drinkwater et al., 1998; Song et al., 2021; Zhang et al., 2023).

The content of non-hydrolysable residue, which characterizes the most resistant part of organic matter to biochemical transformation is of particular interest.

The results suggest that even after a long fallow period, the most stable pool of humus substances has not recovered. Probably, in arid climates, absence of soil cultivating doesn't provide accumulation of stable humus compounds, due to limited input of organic residues and low productivity of natural vegetation. This aligns with findings that in semi-arid conditions, the recovery of organic carbon supplies after stopping cultivation can be slow and doesn't always reach the original level even after 10–15 years without active restoration of plant cover (Chang et al., 2017).

The total content of organic carbon does not reflect the qualitative composition of humic substances or their stability. Therefore, to get a better idea of humus condition, group and fractional composition of humus was analyzed.

The $\Sigma\text{HA}/\Sigma\text{FA}$ ratio of 0.6 indicates the predominance of fulvic acids over humic acids and corresponds to a fulvate-humate type of humus typical of light sierozems in the arid zone. Despite the same $\Sigma\text{HA}/\Sigma\text{FA}$ ratio 0.6, the land-use types differed in the total content and distribution of individual humic substance fractions (Ozhovan, 2013; Skrylnyk et al., 2018).

Fractional composition of humus changed depending on land use type. In soils of three plots, fraction HA_{III} dominated in composition of humic acids, represented by humic substances strongly bound to mineral part of soil and considered more resistant to decomposition (Ponomareva et al., 1980; Orlov, 1990).

The data obtained indicate a higher proportion of organic matter associated with mineral part of soil, which may reflect increase in the processes of forming stable organo-mineral complexes (Alekseeva, 2021; Bezuglova et al., 2024).

The predominance of the FA_{I} fraction indicates a relatively higher proportion of mobile and less stable components of organic matter in the fallow soil. Therefore, simply stopping soil cultivation by itself did not ensure restoration of stable humus pools; in arid conditions, this process is likely limited by poor development of natural vegetation cover and insufficient input of organic residues (Timofeeva et al., 2021; Burdukovskii et al., 2020; Mockeviciene et al., 2022).

Thus, the soil under third-year alfalfa was characterized by a higher humic acid content, a greater proportion of humic substances associated with the soil mineral matrix, and more favorable indicators of humus status. The results show that character of land use affects not only the amount but also the quality of organic matter in light sierozem soils, determining its level of humification and stability.

In irrigated fields, in application of nitrogen two opposing processes are possible. On the one hand, higher crop productivity can increase the input of plant residues and carbon into soil. On the other hand, tillage, irrigation, and nitrogen fertilization can boost mineralization of organic matter, cause losses of soluble organic carbon with drainage water, enhance microbial transformation of organic compounds, and involve previously stabilized forms of organic matter in decomposition processes (Mládková et al., 2006; Zheng et al., 2024). So, soil nitrogen background definitely needs to be considered when studying organic matter transformation in irrigated farming.

In conditions with light granulometric composition and regular irrigation, nitrate nitrogen easily migrates down soil profile, which promotes its downward movement and accumulation in B horizon. At the same time, sources of its accumulation in these areas differ. In arable land, nitrate nitrogen accumulation may be associated with the mineralization of organic matter, use of nitrogen fertilizers and crop residues. The higher nitrate nitrogen content under alfalfa may be indirectly related to symbiotic nitrogen fixation follow by the input, mineralization, and nitrification of nitrogen-containing plant residues (Kebede, 2021).

The minimal content of nitrate nitrogen in all horizons of fallow soil is probably due to the absence of additional nitrogen input and its more active involvement in natural biological cycle.

Thus, under alfalfa and in arable land, nitrate nitrogen increased from the surface horizon to the middle part of the profile and then decreased in the deeper horizon. This pattern should be considered when determining the forms, rates, and timing of nitrogen fertilizer application to avoid unproductive losses.

High content of available phosphorus in topsoil is linked to its low mobility and its binding in carbonate soils by calcium and magnesium. High concentrations of available phosphorus in the upper layers can also be due to accumulation of organic matter and localization of main mass of roots in the plow layer (Hinsinger, 2001; Murad et al., 2024; York et al., 2013; Tang et al., 2023; Peterburgskii, 1981).

Similarly to phosphorus, the higher availability of exchangeable potassium in topsoil is linked to biogenic accumulation of the element, as well as higher levels of fine particles and organic

matter in the upper soil layers. In deeper layers, where the content of clay and organic matter decreases, soil retaining ability in relation to potassium also decreases (Mengel et al., 2001).

One of the indicators of soil agrochemical condition is the content of available sulfur, which plays an important role in nitrogen-use efficiency. The profile distribution of the element showed that the maximum sulfur concentration in the B horizon is typical for soils in areas with perennial alfalfa and arable land, while in fallow area, increase in sulfur content is observed at the depth where parent rock is located.

The profile distribution of available sulfur occupied an intermediate position between those of nitrate nitrogen and available phosphorus.

The higher sulfur content under alfalfa may be related to accumulation of organic matter, activation of microbiological processes, and input of sulfur-containing compounds from residues of perennial crops (Eriksen et al., 1998; Tabatabai, 1996).

At the same time, the increase in available sulfur in the lower horizons shows the high mobility of sulfate ions under conditions of light soil texture and periodic leaching.

The distribution of available nutrients in the soil profile is closely related not only to the particle size composition but also to soil reaction (pH), which affects the mobility, availability, and chemical forms of nitrogen, phosphorus, sulfur, and exchangeable cations.

To characterize the physicochemical state of light sierozems under different land-use conditions, the composition of exchangeable cations in the soil exchange complex was analyzed.

In general, the studied light sierozems differed among land-use types mainly in the profile distribution of calcium and magnesium rather than in sodium content.

CONCLUSION

The studied light sierozem soils were characterized by light loamy texture with predominance of fine sandy particles. The highest physical clay content was found in the fallow soil, and the lowest in the irrigated field under melon crops; the soil under perennial alfalfa occupied an intermediate position. Differences in distribution of fine particles in profile are probably due to combination of natural spatial heterogeneity of soil surface and land use practices.

The most favorable indicators of humus status were found under perennial alfalfa. The organic carbon content here was 0.96%, while in irrigated arable land and fallow plots it was 0.72% and 0.65%, respectively. The soil under alfalfa had the highest content of non-hydrolysable residue, a higher proportion of humic acids, and a larger HA_{III} fraction associated with mineral part of the soil.

In all variants, the ratio $\Sigma\text{HA}/\Sigma\text{FA}$ was 0.6, while fulvic acids were dominating in humus composition. The results indicate that stopping agricultural land use for more than 15 years in arid climate conditions alone did not ensure the restoration of stable organic matter pool.

The distribution of available forms of nutrients showed a clear profile differentiation. Under alfalfa and on irrigated arable land, the maximum content of nitrate nitrogen and available sulfur was observed in the middle part of the profile, indicating redistribution of these elements throughout the profile and their accumulation in the middle part under conditions of irrigation and light soil texture. In contrast, available phosphorus and exchangeable potassium were mainly concentrated in the upper horizons.

On the fallow plot, nitrate nitrogen content was minimal throughout the entire profile.

The soils of all the studied sites were characterized by strongly alkaline reaction and predominance of exchangeable calcium and magnesium. Exchangeable sodium was present in small amounts, indicating a low contribution of sodium to the soil exchange complex.

The main differences between land use types were seen in profile distribution of calcium and magnesium – under alfalfa, calcium content generally decreased with depth; in arable land, it increased in the middle of profile; and in fallow land, there was a noticeable accumulation of both cations in the lower horizons.

Correlation analysis revealed relationships between particle size and agrochemical properties of soils studied. The strongest positive correlation was found between nitrate nitrogen

and available sulfur, while accumulation of available phosphorus and exchangeable potassium was related to the content of physical clay.

The results showed that the soil under third-year alfalfa had a higher organic carbon content and a greater proportion of stable humic substances fractions than the irrigated arable and fallow soils.

To maintain fertility of irrigated light sierozem soils, it is necessary to ensure the supply of organic matter and take into account the possibility of nitrate and sulfate migration when determining forms, doses, and timing of fertilizer application.

At the studies site, leaving irrigated light sierozem soil as unmanaged fallow for more than 15 years did not restore its organic carbon content to the level recorded under alfalfa. When land is temporarily withdrawn from agricultural use, measures are necessary to preserve productive vegetation cover, ensure a regular input of organic matter, and monitor humus status and available nutrient contents.

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AUTHORS CONTRIBUTIONS

Conceptualization of the study, methodology development, project administration: GK; methodology development, interpretation of results, preparation of initial manuscript, reviewing and editing: RR; conducting field and laboratory research, data visualization, preparation of raw data: YB, AT; formal analysis and processing of raw data: AT.

REFERENCES

Soils of the Kazakh SSR. Vol. 14: Kyzylorda Region. 1988. Nauka, Alma-Ata, Kazakhstan [In Russ].

Suleimenov, B.U., Saparov, A.S., Tanirbergenov, S.I. [Current status and prospects for the use of irrigated Sierozem soils in southern Kazakhstan] Sostoyanie i perspektivy ispol'zovaniya oroshaemyh serozemov Yuzhnogo Kazakhstana. Soil Science and Agrochemistry, 2013, 1, 19–27 [In Russ].

Suleimenov, B.U., Saparov, A.S., Dzhalkankuzov, T.D. [Effects of irrigation and fertilization on the chemical properties of irrigated Light Sierozem soils in southern Kazakhstan] Vliyanie orosheniya i udobrenij na himicheskie svoystva oroshaemyh svetlyh serozemov yuzhnogo Kazakhstana. 2009, Soil Science and Agrochemistry, 1, 56–65 [In Russ].

Mamontov, V.G., Efimov, O.E., Kamennykh, N.L. (2023) [Humus status of virgin and cultivated soils of the zonal series: electronic study guide] Gumusnoe sostoyanie tselinnykh i okulturennykh pochv zonalnogo ryada: elektronnoe uchebnoe posobie. Moscow: Russian State Agrarian University–Moscow Timiryazev Agricultural Academy. [124088_8f573d0d12f515abbf2ff6586907d3bd.pdf](https://doi.org/10.124088_8f573d0d12f515abbf2ff6586907d3bd.pdf) [In Russ]

Orlov, D.S. [Soil chemistry] Khimiya pochv. Moscow: Moscow State University Press, 1992, 400 p. [In Russ]

Vorobyeva, L.A. [Chemical analysis of soils] Khimicheskii analiz pochv. Moscow: Moscow State University Press, 1998, 272 p. [In Russ]

Borovsky, V.M., Uspanov, U.U. [Soils of Kazakhstan and their economic utilization] Pochvy Kazakhstana i ikh khozyaistvennoe ispol'zovanie. Alma-Ata: Academy of Sciences of the Kazakh SSR, 1971. 25 p. [In Russ]

Zhou, Z., Zhang, X., Gan, Z. (2015). Changes in soil organic carbon and nitrogen after 26 years of farmland management on the Loess Plateau of China. Journal of Arid Land 7(6), 806–813. <https://doi.org/10.1007/s40333-015-0051-y>

- Jensen, E.S., Peoples, M.B., Boddey, R.M., Gresshoff, P.M., Hauggaard-Nielsen, H., Alves, B.J.R., Morrison, M.J. (2012). Legumes for mitigation of climate change and the provision of feedstock for biofuels and biorefineries: A review. *Agronomy for Sustainable Development*, 32, 329–364. <https://doi.org/10.1007/s13593-011-0056-7>
- Drinkwater, L.E., Wagoner, P., Sarrantonio, M. (1998). Legume-based cropping systems have reduced carbon and nitrogen losses. *Nature*, 396, 262–265. <https://doi.org/10.1038/24376>
- Chang, X., Chai, Q., Wu, G., Zhu, Y., Li, Z., Yang, Y., Wang, G. (2017). Soil organic carbon accumulation in abandoned croplands on the Loess Plateau. *Land Degradation & Development*, 28, 1519–1527. <https://doi.org/10.1002/ldr.2679>
- Zhang, B., Gao, Z., Zhi, J., Bai, X., Yang, L., Xia, W. (2023). Effects of irrigation and nitrogen fertilizer on soil carbon leaching in cotton fields in arid areas. *Sustainability*, 15, 11356. <https://doi.org/10.3390/su151411356>
- Song, X., Fang, C., Yuan, Z.-Q., Li, F.-M. (2021). Long-term growth of alfalfa increased soil organic matter accumulation and nutrient mineralization in a semi-arid environment. *Frontiers in Environmental Science*, 9:649346. <https://doi.org/10.3389/fenvs.2021.649346>
- Ozhovan, E.A. Humus status and buffer properties of Chernozems in southwestern Ukraine [Gumusnoe sostoyanie i bufernye svojstva chernozemov yugo-zapada Ukrainy] *Soil Science and Agrochemistry*, 2013, 2(51), 102–112 [In Russ].
- Skrylnyk, E.V., Tovsty, Y.N. [Changes in humus content and quality in podzolized Chernozem after application of poultry manure and composts] *Izmenenie sodержaniya i kachestva gumusa v chernozeme opodzolennom posle vneseniya pometa i kompostov na ego osnove*, *Bulletin of Karaganda University. Biology, Medicine, Geography Series*, 2018, 4(92), 17–25 [In Russ].
- Ponomareva, V.V., Plotnikova, T.A. [Humus and Soil Formation (Methods and Research Results)] *Gumus i pochvoobrazovanie (metody i rezul'taty izucheniya)*, 1980, Nauka, Leningrad [In Russ].
- Orlov, D.S. *Humic Substances of Soils and the General Theory of Humification*. Moscow: Moscow State University Press, 1990, 266.
- Alekseeva, T.V. [Stable soil organic matter: Characteristics and fate during diagenesis] *Prochnosvyazannoe organicheskoe veshchestvo pochv: harakteristika i sud'ba v diageneze*. *Soils and Environment*, 2021, 4(4), e161. <https://doi.org/10.31251/pos.v4i4.161> [In Russ]
- Bezuglova, O.S., Kushnareva, A.V., Omelyanchuk, A.A., Tishchenko, S.A. (2024). [Humus state of Dark Chestnut soils derived from rice crop rotation]. *Gumusnoe sostoyanie temno-kashtanovykh pochv, vyvedennykh iz risovogo sevooborota*, [st_646.pdf](#) [In Russ]
- Timofeeva, Y., Purtova, L., Emelyanov, A., Burdukovskii, M., Kiseleva, I., Sidorenko, M. (2021). Contents, distribution, and fractionation of soil organic carbon and trace elements in soils under green manure application. *Soil & Water Research*, 16, 50–58. <https://doi.org/10.17221/65/2020-SWR>
- Burdukovskii, M., Kiseleva, I., Perepelkina, P., Kosheleva, Y. (2020). Impact of different fallow durations on soil aggregate structure and humus status parameters. *Soil & Water Research* 15, 1–8. <https://doi.org/10.17221/174/2018-SWR>
- Mockeviciene, I., Karcauskiene, D., Slepeticene, A., Vilkiene, M., Repsiene, R., Braziene, Z., Anne, O., 2022. Influence of liming intensity on fractions of humified organic carbon in acid soil: A case study. *Sustainability* 14, 5297. <https://doi.org/10.3390/su14095297>
- Mládková, L., Rohošková, M., Borůvka, L. (2006). Methods for the assessment of humic substances quality in forest soils. *Soil & Water Research*, 1, 3–9. <https://doi.org/10.17221/6499-SWR>
- Zheng, Y., Jin, J., Wang, X., Kopittke, P.M., O'Sullivan, J.B., Tang, C. (2024). Disentangling the effect of nitrogen supply on the priming of soil organic matter: A critical review. *Critical Reviews in Environmental Science and Technology* 54, 1118–1152. <https://doi.org/10.1080/10643389.2023.2266312>

Kebede, E. (2021). Contribution, utilization, and improvement of legumes-driven biological nitrogen fixation in agricultural systems. *Frontiers in Sustainable Food Systems*, 5, 767998. <https://doi.org/10.3389/fsufs.2021.767998>

Hinsinger, P. (2001). Bioavailability of soil inorganic phosphorus in the rhizosphere as affected by root-induced chemical changes: A review. *Plant and Soil* 237, 173–195. <https://doi.org/10.1023/A:1013351617532>

Murad, S., Ahmad, M., Hussain, A., Ali, S., Al-Ansari, N., A.Mattar, M. (2024). Efficacy of DAP coated with bacterial strains and their metabolites for soil phosphorus availability and maize growth. *Scientific Reports*, 14, 11389. <https://doi.org/10.1038/s41598-024-61817-6>

York, L.M., Nord, E.A., Lynch, J.P. (2013). Integration of root phenes for soil resource acquisition. *Frontiers in Plant Science*, 4:355. <https://doi.org/10.3389/fpls.2013.00355>

Tang, X., Liu, H., Qin, H., Zhao, J., Wang, H., Li, B., Lu, Y. (2023). Organic/inorganic phosphorus partition and transformation during long-term paddy cultivation in the Pearl River Delta, China. *Scientific Reports*, 13:11122. <https://doi.org/10.1038/s41598-023-38369-2>

Peterburgskii, A.V. *Agrochemistry and Plant Nutrition Physiology*. Moscow: Rosselkhozizdat, 1981 333 p. [In Russ].

Mengel, K., Kirkby, E.A. (2001). *Principles of Plant Nutrition*, 5th ed. Kluwer Academic Publishers, Dordrecht.

Eriksen, J., Murphy, M.D., Schnug, E. (1998). The Soil Sulphur Cycle. In: Schnug, E. (eds) *Sulphur in Agroecosystems. Nutrients in Ecosystems*, vol 2. Springer, Dordrecht. https://doi.org/10.1007/978-94-011-5100-9_2

Tabatabai, M.A. (1996). Sulfur. In *Methods of Soil Analysis* (eds D.L. Sparks, A.L. Page, P.A. Helmke, R.H. Loeppert, P.N. Soltanpour, M.A. Tabatabai, C.T. Johnston and M.E. Sumner). <https://doi.org/10.2136/sssabookser5.3.c33>

ТҮЙІН

Ашық боз топырақтар Қазақстанның оңтүстігіндегі суармалы егіншілікте кеңінен пайдаланылады, алайда органикалық заттар мөлшерінің төмендігі және екіншілік тұздануға бейімділігі олардың құнарлылығын жерді пайдалану сипатына тәуелді етеді. Зерттеудің мақсаты Қызылорда облысындағы жерді пайдаланудың қарама-қарсы түрлеріндегі суармалы ашық боз топырақтардың қарашірінді жағдайы мен агрохимиялық қасиеттерін салыстырмалы бағалау болды.

Үшінші жыл өсіріліп жатқан жоңышқа танабында, бақша дақылдары егілген суармалы егістікте және 15 жылдан астам уақыт ауыл шаруашылығында пайдаланылмаған танаптарда салынған үш топырақ кескіні зерттелді. Үлгілер тектік қабаттар бойынша алынып, олардың гранулометриялық құрамы, топырақтың органикалық көміртегі (SOC), қарашірінді фракциялары, өсімдікке қолжетімді қоректік элементтер, рН және алмаспалы катиондар анықталды.

Барлық топырақтар ұсақ құм фракциясы басым жеңіл құмбалшықты гранулометриялық құраммен сипатталды. Жоғарғы қабаттағы SOC мөлшері жоңышқа танабында 0,96%, суармалы егістікте 0,72%, ал тыңайған топырақта 0,65%. Жоңышқа егілген топырақта гидролизденбейтін қалдық пен гумин қышқылдарының мөлшері жоғары болды. Барлық топырақтарда ΣНА/ΣFA қатынасы 0,6 құрады. Жоңышқа танабы мен суармалы егістікте нитратты азот пен өсімдікке қолжетімді күкірт топырақ кескінінің ортаңғы бөлігінде жиналды, ал өсімдікке қолжетімді фосфор мен алмаспалы калий негізінен жоғарғы қабаттарда шоғырланған. Топырақтың рН көрсеткіші 8,76–9,07 аралығында.

Үшінші жыл өсіріліп жатқан жоңышқа танабының топырағында SOC мөлшері және қарашіріндінің тұрақты фракцияларының үлесі ең жоғары мәнге ие болды. Зерттелген танапта 15 жылдан астам уақыт бойы басқарылмай тыңайған күйде қалдыру органикалық заттардың салыстырмалы тұрақты қорын қалпына келтірмеді.

РЕЗЮМЕ

Светлые серозёмы широко используются в орошаемом земледелии юга Казахстана, но низкое содержание органического вещества и риски проявления вторичного засоления в зависимости от характера землепользования значительно влияют на плодородия этих почв. Цель исследования заключалась в сравнительной оценке гумусного состояния и агрохимических свойств орошаемых светлых серозёмов при контрастных типах землепользования в Кызылординской области.

Исследованы три почвенных разреза, заложенных на поле люцерны третьего года жизни, орошаемой пашне под бахчевыми культурами и участке, не использовавшемся в сельском хозяйстве более 15 лет. Образцы отбирали по генетическим горизонтам, определяли гранулометрический состав, содержание органического углерода почвы (SOC), фракционный состав гумуса, подвижные формы элементов питания, pH и состав обменных катионов.

Для всех почв характерным являлся легкосуглинистый гранулометрический состав с преобладанием мелкопесчаной фракции. Содержание SOC в верхнем горизонте составляло 0,96% под люцерной, 0,72% на орошаемой пашне и 0,65% в залежной почве. Почва под люцерной содержала больше негидролизуемого остатка и гуминовых кислот. Во всех почвах отношение $\Sigma\text{HA}/\Sigma\text{FA}$ составляло 0,6. Нитратный азот и подвижная сера накапливались в средней части профиля под люцерной и на орошаемой пашне, тогда как доступный фосфор и обменный калий концентрировались преимущественно в верхних горизонтах. Значения pH находились в пределах 8,76–9,07.

Почва под люцерной третьего года жизни характеризовалась наибольшим содержанием SOC и более высокой долей устойчивых фракций гумуса. На обследованном участке более чем 15-летнее залежное состояние не обеспечило восстановления сопоставимого устойчивого пула органического вещества.

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